Community Solar Credit Allocation Manager

Requirements Table | Lab 1 | Problem Statement #63

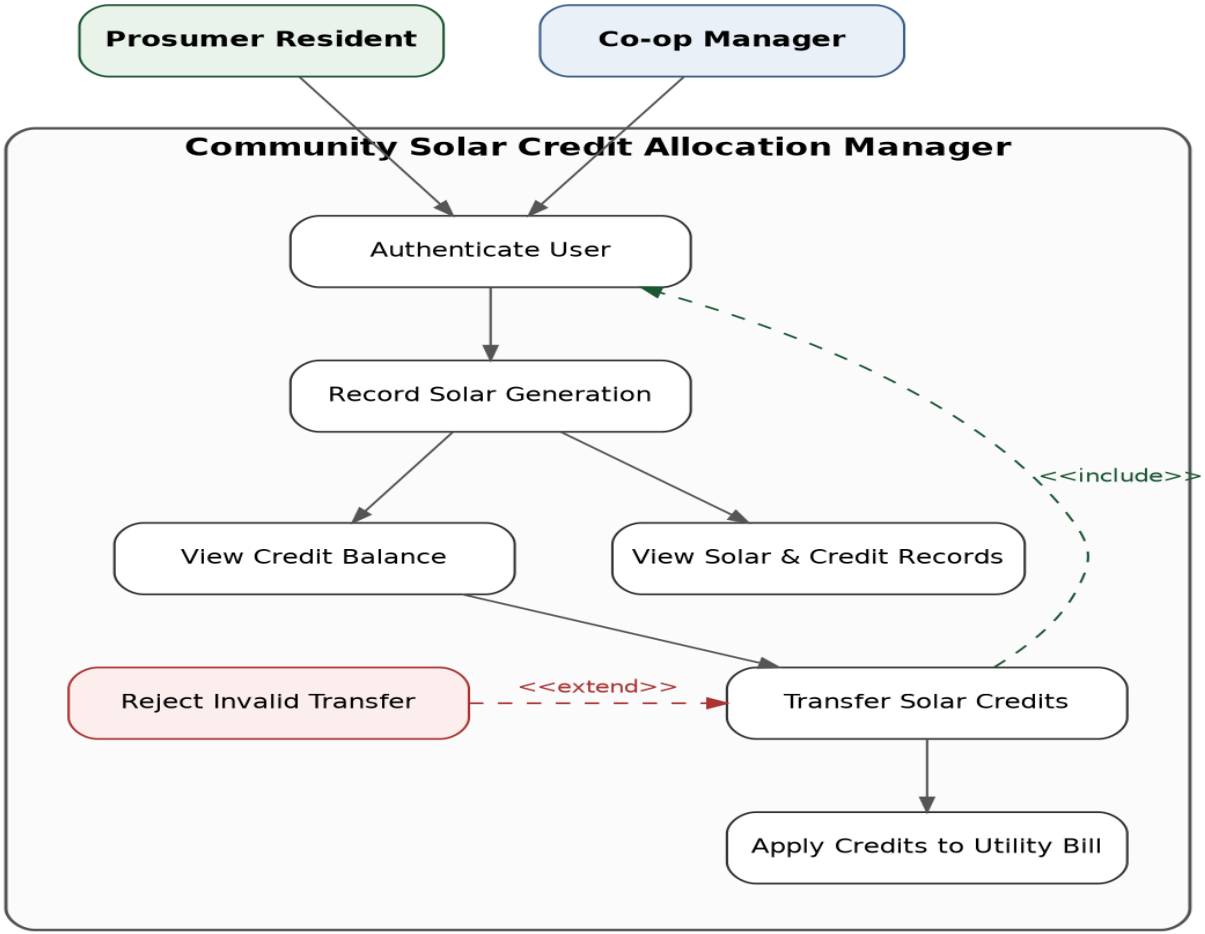
Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall record solar generation data in kWh received from smart meters for each prosumer resident.	High	Pass: Valid generation data is stored with date/time and resident ID. Fail: Invalid or incomplete meter data is rejected.	Accurate generation records are required to calculate available solar credits.
FR-002	Functional	The system shall calculate and display the available solar credit balance for each prosumer resident.	High	Pass: Credit balance reflects recorded generation and previous transfers. Fail: Balance is incorrect or unavailable.	Residents need to know how many credits they can use or transfer.
FR-003	Functional	The system shall allow a prosumer resident to transfer surplus solar credits to another resident's account.	High	Pass: Credits are deducted from the sender and added to the recipient. Fail: Transfer exceeding available balance is rejected.	Enables peer-to-peer sharing of surplus solar credits.
FR-004	Functional	The system shall apply received solar credits to a resident's monthly utility billing statement.	High	Pass: Valid credits reduce the resident's billing amount. Fail: Invalid or unavailable credits are not applied.	The main benefit of the system is reducing utility bills using shared credits.
FR-005	Functional	The system shall allow the co-op manager to view solar generation, credit transfers, and billing-offset records.	Medium	Pass: Manager can view records for selected residents and periods. Fail: Unauthorized users cannot access management records.	Helps the co-op manager monitor and manage community solar activity.

Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance	The system shall process and store hourly solar-generation data for at least 500 households while maintaining 99.9% availability.	High	Pass: Load testing confirms successful processing for 500 households and 99.9% availability. Fail: The system cannot meet the required load or availability.	Reliable processing is necessary for accurate credit allocation and billing.
NFR-002	Security	The system shall authenticate users and protect solar-generation, credit-transfer, and billing information from unauthorized access.	High	Pass: Only authenticated and authorized users can access protected functions and data. Fail: An unauthorized user gains access.	Solar and billing information is sensitive and must be protected.

Use-Case Diagram:-



Use-Case Flow Specification

Use Case ID	UC-01
Use Case Name	Transfer Solar Credits
Primary Actor	Prosumer Resident
Secondary Actor(s)	Co-op Manager (indirect)

Preconditions

- The prosumer resident has a registered account.
- The prosumer is authenticated (see <<include>> Authenticate User).
- The prosumer has a non-zero available solar credit balance.
- The recipient resident has a valid, active account in the system.

Postconditions

- The transferred credits are deducted from the sender's balance.
- The transferred credits are added to the recipient's balance.
- The transfer is recorded in the system's transaction history with a timestamp.
- The updated balances are immediately available to both users.

Main Success Scenario

1. The prosumer resident logs into the system.
2. The system authenticates the prosumer (<<include>> Authenticate User).
3. The prosumer selects Transfer Solar Credits.
4. The system displays the prosumer's available credit balance.
5. The prosumer enters the recipient's account identifier and the number of credits to transfer.
6. The system validates the recipient account and the requested transfer amount.
7. The system deducts the specified credits from the sender's account.
8. The system adds the transferred credits to the recipient's account.
9. The system records the transaction in the transfer history.
10. The system displays a successful-transfer confirmation to the prosumer.

Alternate Flow – A1: Insufficient Credit Balance

Triggered at Step 6, when the requested transfer amount exceeds the sender's available balance

- 6a. The system determines that the sender does not have enough credits to complete the transfer.
- 6b. The system rejects the transfer request.
- 6c. The system displays the message: "Transfer failed: Insufficient solar credit balance."
- 6d. No credits are deducted from the sender or added to the recipient.
- 6e. The use case resumes at Step 5, allowing the prosumer to re-enter a valid transfer amount.